Individual Contribution Report

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Group: Shared Brain cells

Assigned Role: File Management Developer

What did I contribute to the project?

I Built File system.py, and other commands like write, read, delete and CD commands. And in the Documentation

What features did I personally develop?

What I developed is making commands inside the OS to read, run, process, and others.

What problems did I encounter? Sometimes I don't see the command results for the actions I've performed; they just don't show up.

How did I solve those problems?

The logic and lines were double-checked for errors and tested repeatedly to ensure a high-quality outcome.

What did I learn from this project? One of the things I learned is the importance of patience in working, in thinking, in approaching problems, and in collaborating within a group.

Evidence of my contribution:



Code Blame 74 lines (64 loc) · 2.07 KB

```python
class FileSystem:
    """Simulated file system. Folders are dicts. Files are strings."""

    def __init__(self):
        self.root = {}
        self.path = []

    def _cwd(self):
        node = self.root
        for part in self.path:
            node = node[part]
        return node

    def location(self):
        return "/" + "/".join(self.path)

    def dir(self):
        items = self._cwd()
        if not items:
            return ["(empty)"]
        lines = []
        for name, value in sorted(items.items()):
            if isinstance(value, dict):
                lines.append(f"<DIR>   {name}")
            else:
                lines.append(f"        {name}")
        return lines

    def mkdir(self, name):
        cwd = self._cwd()
        if name in cwd:
            return "Error: Name already exists."
        cwd[name] = {}
        return "Directory created."

    def create(self, name):
        cwd = self._cwd()
        if name in cwd:
            return "Error: Name already exists."
        cwd[name] = ""
        return "File created."

    def write(self, name, text):
        cwd = self._cwd()
        if name not in cwd or isinstance(cwd[name], dict):
            return "Error: File does not exist."
        cwd[name] = text
        return "File saved."

    def read(self, name):
        cwd = self._cwd()
        if name not in cwd or isinstance(cwd[name], dict):
            return "Error: File does not exist."
        return cwd[name] if cwd[name] else "(file is empty)"

    def delete(self, name):
        cwd = self._cwd()
```

```
1    class FileSystem:
17       def dir(self):
21           lines = []
22           for name, value in sorted(items.items()):
23               if isinstance(value, dict):
24                   lines.append(f"<DIR>   {name}")
25               else:
26                   lines.append(f"        {name}")
27           return lines
28
29       def mkdir(self, name):
30           cwd = self._cwd()
31           if name in cwd:
32               return "Error: Name already exists."
33           cwd[name] = {}
34           return "Directory created."
35
36       def create(self, name):
37           cwd = self._cwd()
38           if name in cwd:
39               return "Error: Name already exists."
40           cwd[name] = ""
41           return "File created."
42
43       def write(self, name, text):
44           cwd = self._cwd()
45           if name not in cwd or isinstance(cwd[name], dict):
46               return "Error: File does not exist."
47           cwd[name] = text
48           return "File saved."
49
50       def read(self, name):
51           cwd = self._cwd()
52           if name not in cwd or isinstance(cwd[name], dict):
53               return "Error: File does not exist."
54           return cwd[name] if cwd[name] else "(file is empty)"
55
56       def delete(self, name):
57           cwd = self._cwd()
58           if name not in cwd:
59               return "Error: File does not exist."
60           if isinstance(cwd[name], dict) and cwd[name]:
61               return "Error: Directory is not empty."
62           del cwd[name]
63           return "File deleted."
64
65       def cd(self, name):
66           if name == "..":
67               if self.path:
68                   self.path.pop()
69               return ""
70           cwd = self._cwd()
71           if name not in cwd or not isinstance(cwd[name], dict):
72               return "Error: Directory does not exist."
73           self.path.append(name)
74           return ""
```